Day 072 | Unit 05 | Teacher plan

Represent and Solve Equations and Inequalities

Identify Independent and Dependent Variables

Standards: 6.EE.C.9

Learning target: I can identify quantities that change together and write a rule.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. The independent variable is the input you choose; the dependent variable changes with it. Define both and write the output in terms of the input. A table and graph must show the same rule and units.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 5

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01 / Learn

The independent variable is the input you choose; the dependent variable changes with it. Define both and write the output in terms of the input. A table and graph must show the same rule and units.

Worked example 1

A shop charges $7 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

1. The independent variable is the input you choose; the dependent variable changes with it.

2. Define both and write the output in terms of the input.

3. The chosen notebook count is the input

4. multiply it by price and add any fixed fee to get the cost.

5. State the result in the form requested: Independent n = notebooks; dependent C = cost; C = 7n.

Answer: Independent n = notebooks; dependent C = cost; C = 7n

Worked example 2

A machine uses 4 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (3, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.

1. The independent variable is the input you choose; the dependent variable changes with it.

2. Hours are the input.

3. Multiply each hour value by 4 to obtain liters.

4. Graph hours horizontally and liters vertically.

Answer: (0,0), (1,4), (3,12); independent h; L = 4h

Day 072 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A shop charges $5 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

02 / Constructed response

A shop charges $4 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

03 / Constructed response

A shop charges $3 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

04 / Constructed response

A shop charges $2 for each of n notebooks plus a single $3 delivery fee. Name the independent and dependent variables and write a cost rule.

05 / Constructed response

A shop charges $2 for each of n notebooks plus a single $4 delivery fee. Name the independent and dependent variables and write a cost rule.

06 / Constructed response

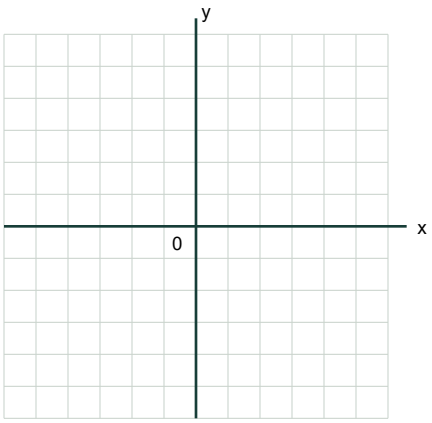
A shop charges $8 for each of n notebooks plus a single $2 delivery fee. Name the independent and dependent variables and write a cost rule.

07 / Constructed response

A shop charges $4 for each of n notebooks plus a single $3 delivery fee. Name the independent and dependent variables and write a cost rule. What does the cost at n = 0 mean in this model?

08 / Table completion, equation, and graph

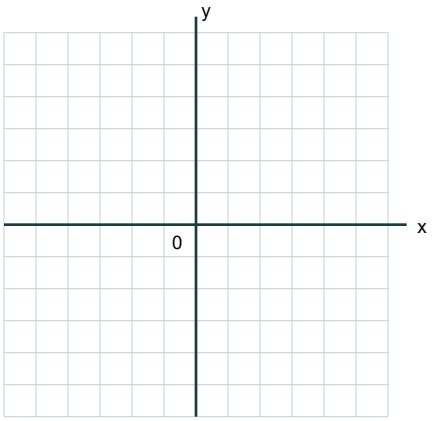
A machine uses 6 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (4, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

09 / Table completion, equation, and graph

A machine uses 3 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (4, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

10 / Constructed response

A shop charges $5 for each of n notebooks plus a single $3 delivery fee. Name the independent and dependent variables and write a cost rule. What does the cost at n = 0 mean in this model?

Day 072 | Unit 05 | Answer key

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1. Independent n = notebooks; dependent C = cost; C = 5n

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

2. Independent n = notebooks; dependent C = cost; C = 4n

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

3. Independent n = notebooks; dependent C = cost; C = 3n

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

4. Independent n = notebooks; dependent C = cost; C = 2n + 3

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

5. Independent n = notebooks; dependent C = cost; C = 2n + 4

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

6. Independent n = notebooks; dependent C = cost; C = 8n + 2

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

7. Independent n = notebooks; dependent C = cost; C = 4n + 3; C(0) = $3, the fee

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

8. (0,0), (1,6), (4,24); independent h; L = 6h

Hours are the input. Multiply each hour value by 6 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

9. (0,0), (1,3), (4,12); independent h; L = 3h

Hours are the input. Multiply each hour value by 3 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

10. Independent n = notebooks; dependent C = cost; C = 5n + 3; C(0) = $3, the fee

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response